PART 3 · SHEET 3.1-1

RAM READER

3.1.1 INTRODUCTION.

The RAM reader is System One's eye. It reads the emulator memory directly: map, player tile, facing, party, collision, and the sprite state table. It renders the room as a 10 by 9 tile grid with the player's facing arrow in his own cell (figure 3.1-1).

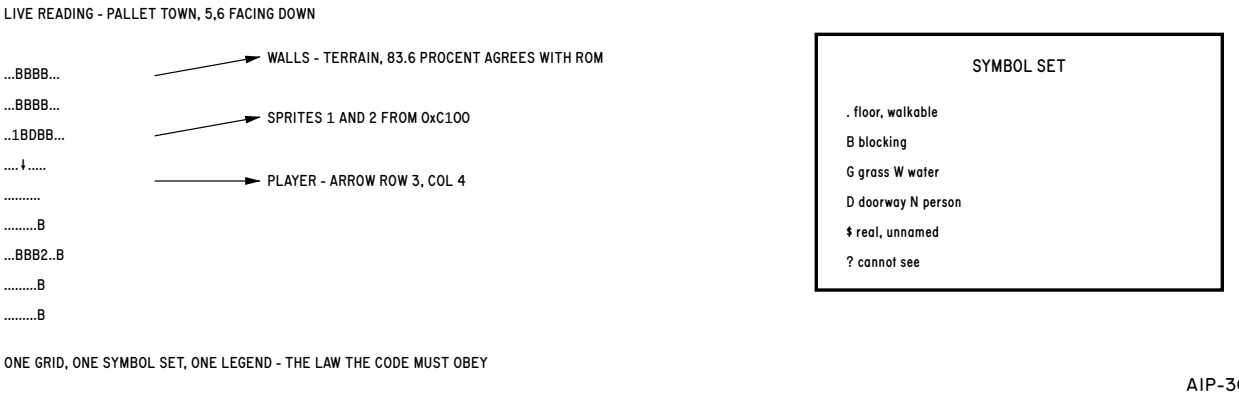


Figure 3.1-1. The Live Tile Grid And Its Legend, With Measured Cells

3.1.2 FUNCTIONAL DESCRIPTION.

Terrain comes from the ROM tile table; sprites come from wSpriteStateData1 at address C100, sixteen records of sixteen bytes, the player in slot zero. Facing renders as an arrow. The grid is the reference the vision channel is judged against, and its own terrain labels measured 83.6 percent against ROM collision.

3.1.2.1 Reference, Not Truth.

The reader is a reference, not an oracle. Sprite meaning, dialogue and menus are invisible to it. A verdict disagreement is a finding to investigate, never a reason to overwrite one channel with the other.

3.1.2.2 What The Reader Cannot See.

Text, menus, dialogue and name entry are invisible to RAM. A sprite's meaning is not in memory either. An out-of-range cell answers walkable, so read the code before trusting a verdict row that says the way is clear.

3.1.2.3 Cost.

The read is free and runs every cycle, which is exactly why it is the reference channel and the vision model is the guest that must justify its 25,000 tokens.

3.1.3 PERFORMANCE AND DESIGN DATA.

Table 3.1-1. Reader Data

Parameter	Value
Grid	10 cols x 9 rows, arrow in player cell
Cell	16 px, origin measured on screen
Sprite table	0xC100, 16 x 16 bytes
Player slot	0, excluded from people probes
Terrain vs ROM	83.6 percent agreement
Cost per read	free, every cycle

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TAKE A READING                    OPERATOR
1 BRIDGE - verify health
2 STATE - read /api/state
  √ GRID - 9 rows of 10
4 ARROW - confirm facing
```

RAM